

ECE435: Medical Imaging Project 1

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Introduction

Image processing is essential to the field of medical imaging. In this report, various image processing techniques are applied to a real volume scan – a CT scan of the thorax from Mulrecon [1]. These concepts and techniques implemented in MATLAB include: noise, filtering, blurring, and edge detection.

Part I: Slicing

The volume scan downloaded from [1] include $237 \times 512 \times 512$ JPG images. These JPG images were then converted into grayscale using `im2gray` and stacked into a $512 \times 512 \times 237$ 3-dimensional tensor. The first dimension represents the x-direction pixel, the second dimension represents the y-direction pixel, and the third dimension refers the stack image. Each x-y pixel corresponds to a length of 0.703 mm, and the spacing between each image (z-direction) is 0.625 mm.

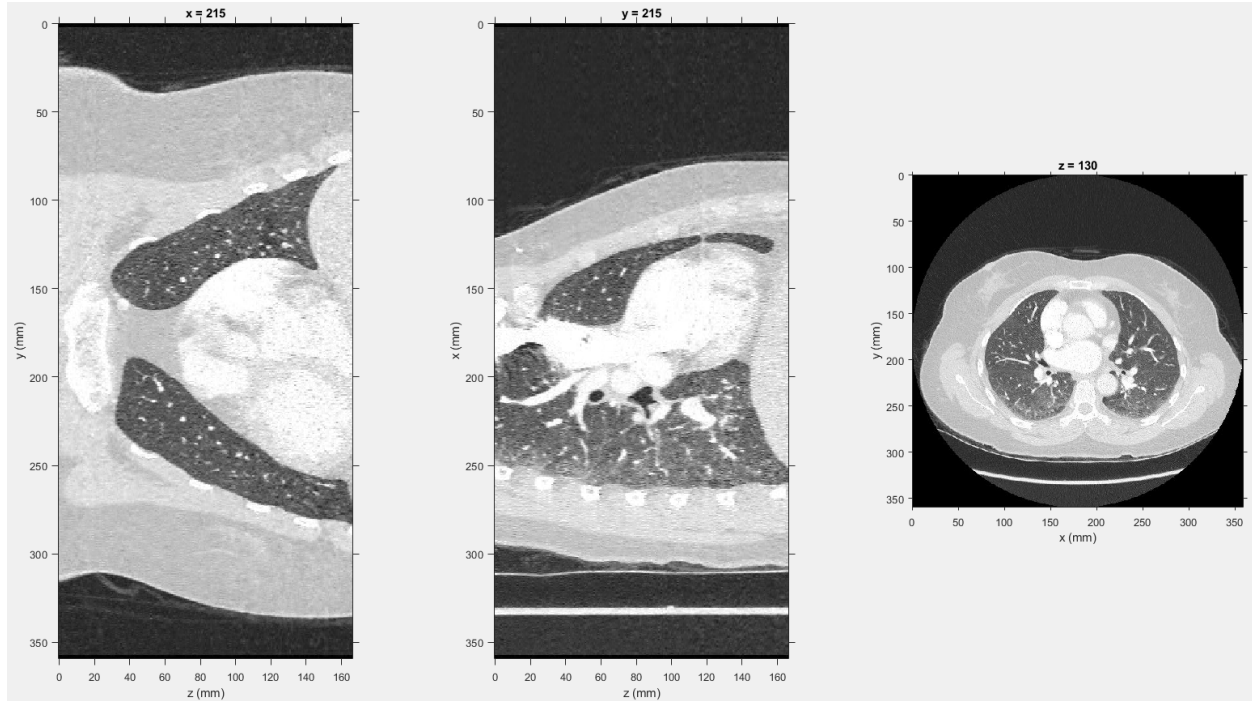


Figure 1: The dimensions of the images (from left to right) are 237×512 , 237×512 , and 512×512 pixels. Equivalently, the images are 148.125mm x 359.936mm, 148.125mm x 359.936mm, and 359.936mm x 359.936mm.

Part II: Salt and Pepper

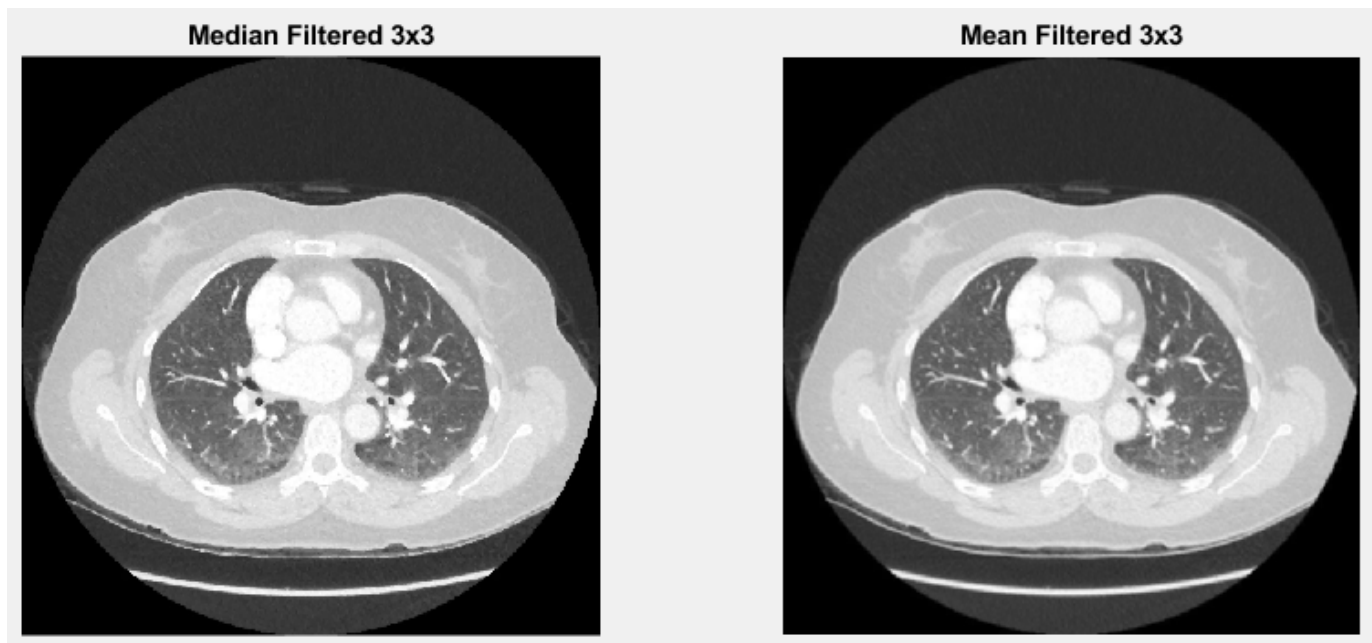


Figure 2: 3x3 Median and Mean filters applied to slice $z = 130$.

Both the mean and median filtered images for both 3x3 and 5x5 filters appear with less grainy compared to the raw images. However, the larger filter size smooths out the image more, making it appear blurrier compared to the smaller filter size. In the case of the 3x3 filter, both the median and mean filtered images look similar, but some areas look do look darker in the median filtered image.

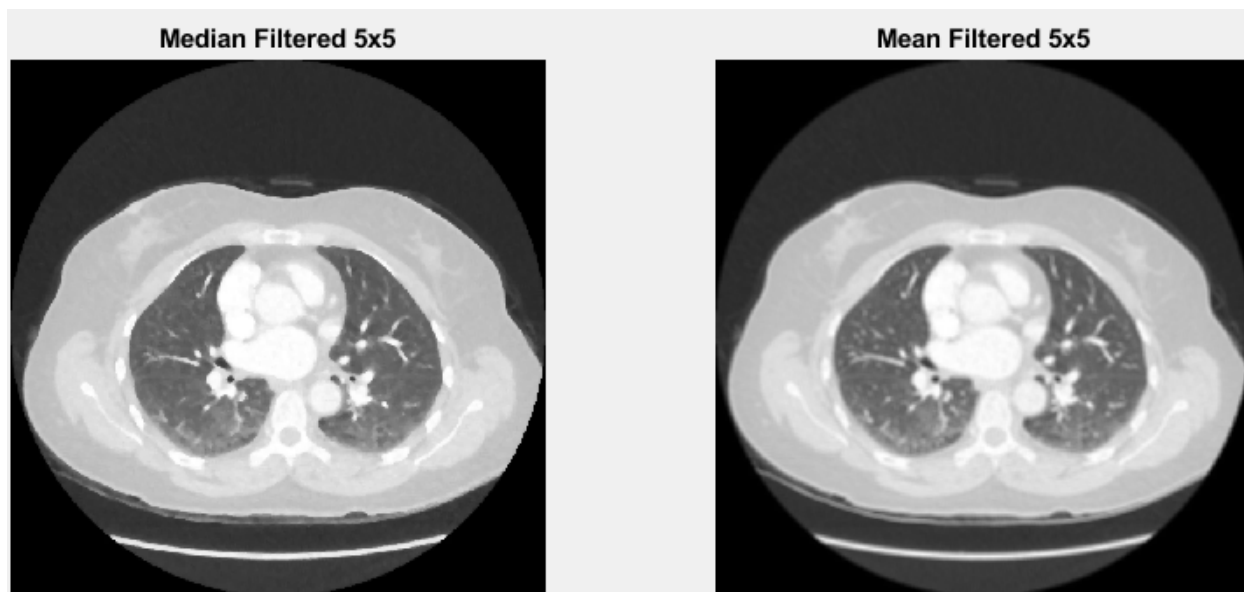


Figure 3: 5x5 Median and Mean filters applied to slice $z = 130$.

After applying Salt-n-Pepper noise using the `imnoise` function, the images look noticeably worse, with random specks spread throughout the image (as expected).

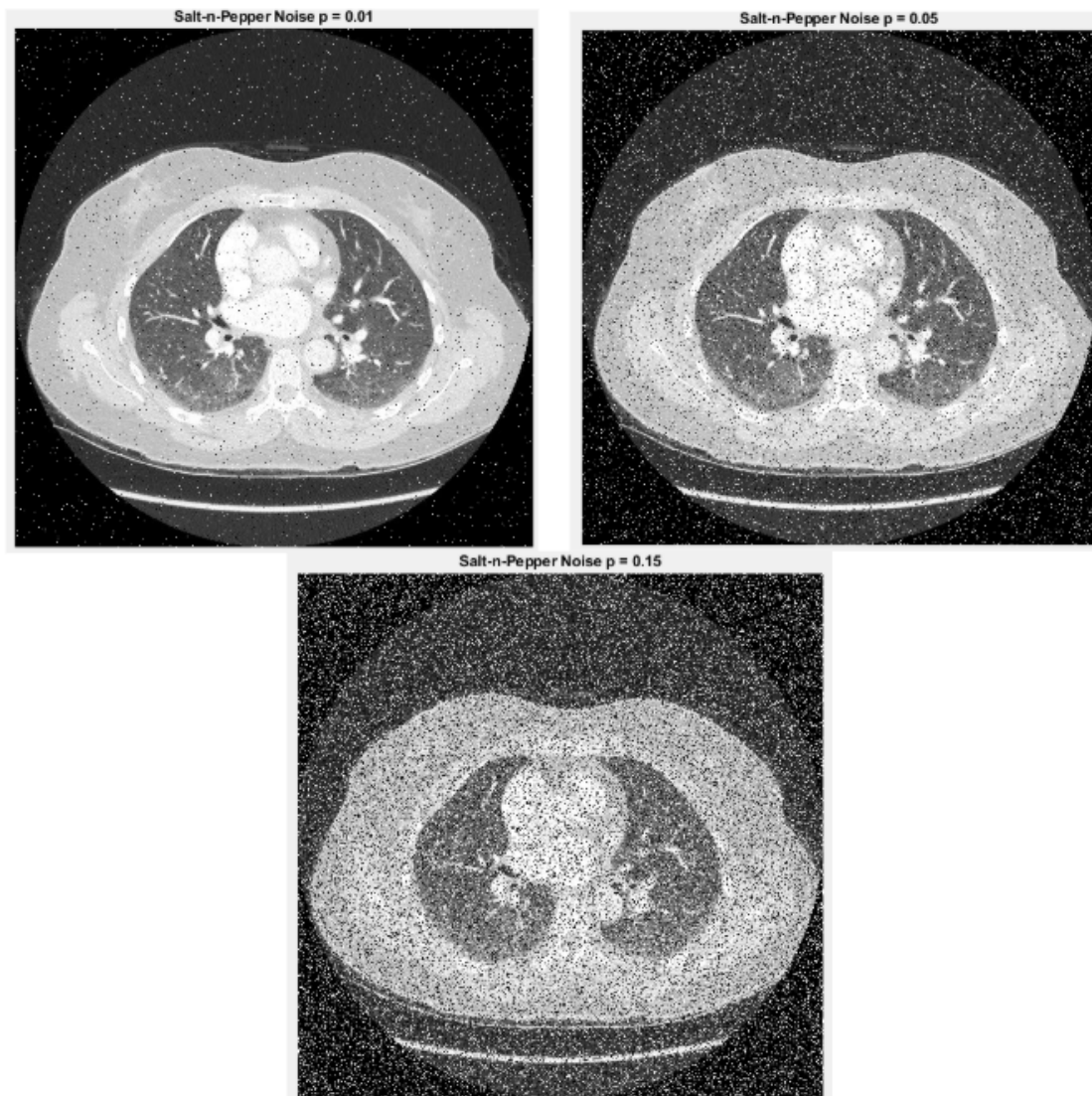


Figure 4: Salt-n-Pepper Noise of various probabilities added to slice $z = 130$.

Applying the 3×3 mean filter to these noisy images does not effectively remove the noise. In the noise $p = 0.01$ image, there are still grey specks throughout the image. This effect is even more exaggerated in the noise $p = 0.15$ image where the image is extremely grainy and hard to make out any finer details besides general outlines. Salt-n-pepper noise is impulsive and sparse, consisting of extreme-valued outliers $\{0, 255\}$. Since the mean filter is not robust to outliers, these extreme values bias the local average instead of being eliminated, resulting in the grainy images seen.

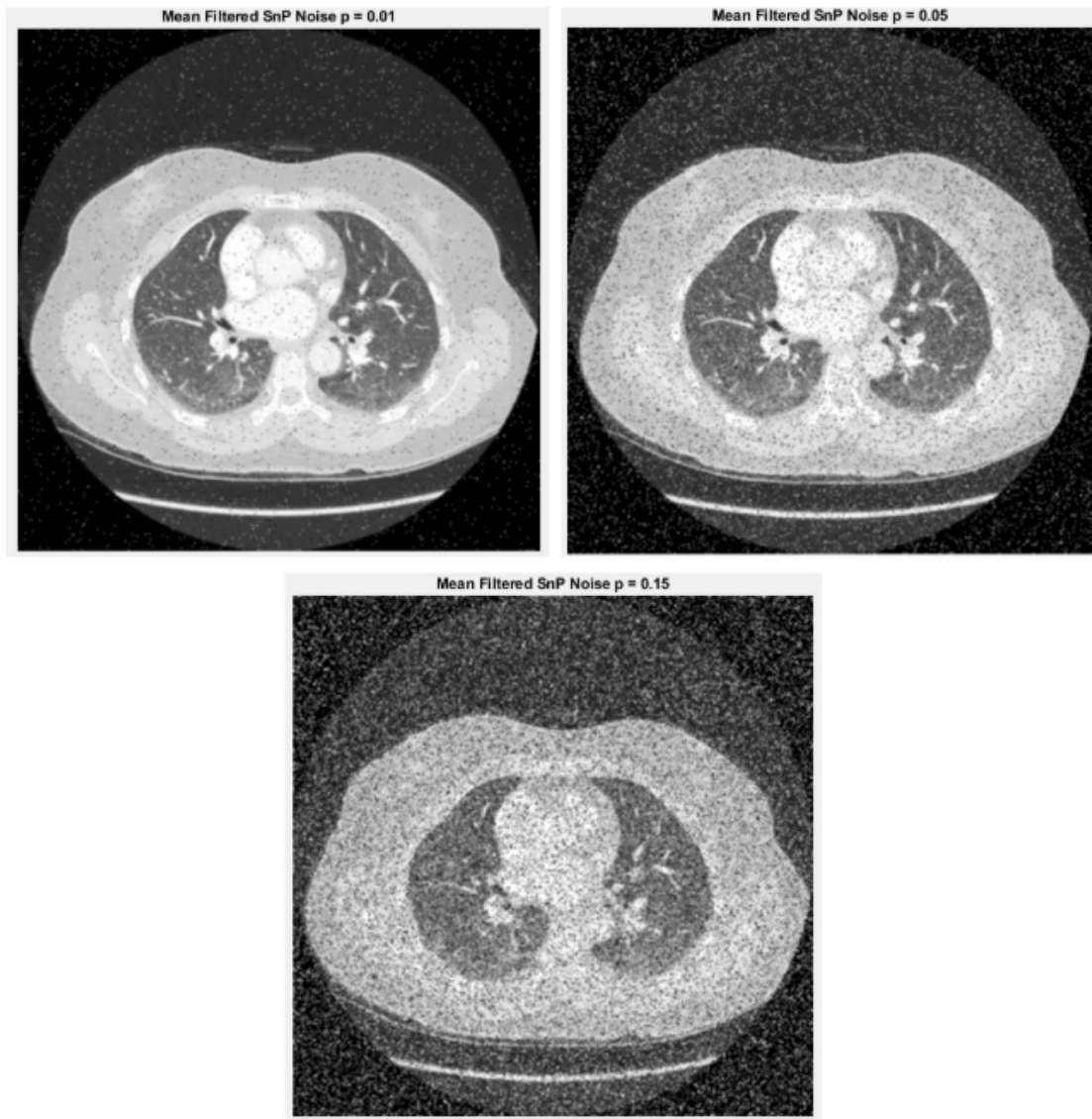


Figure 5: 3x3 Mean Filtered Salt-N-Pepper Noise image slice $z = 130$.

On the contrary, median filters are much more robust to statistical outliers. This effect can be seen in the resulting images. In the noise $p = 0.01$ and $p = 0.05$ images, essentially all the specks are removed. This leads to a clear image. Even when the noise has been increased to $p = 0.15$, the median filter does a substantially better job compared to the mean filter. The image is less grainy and although there are still visible specks of black and white throughout the image, the finer details of the original image are still visible.

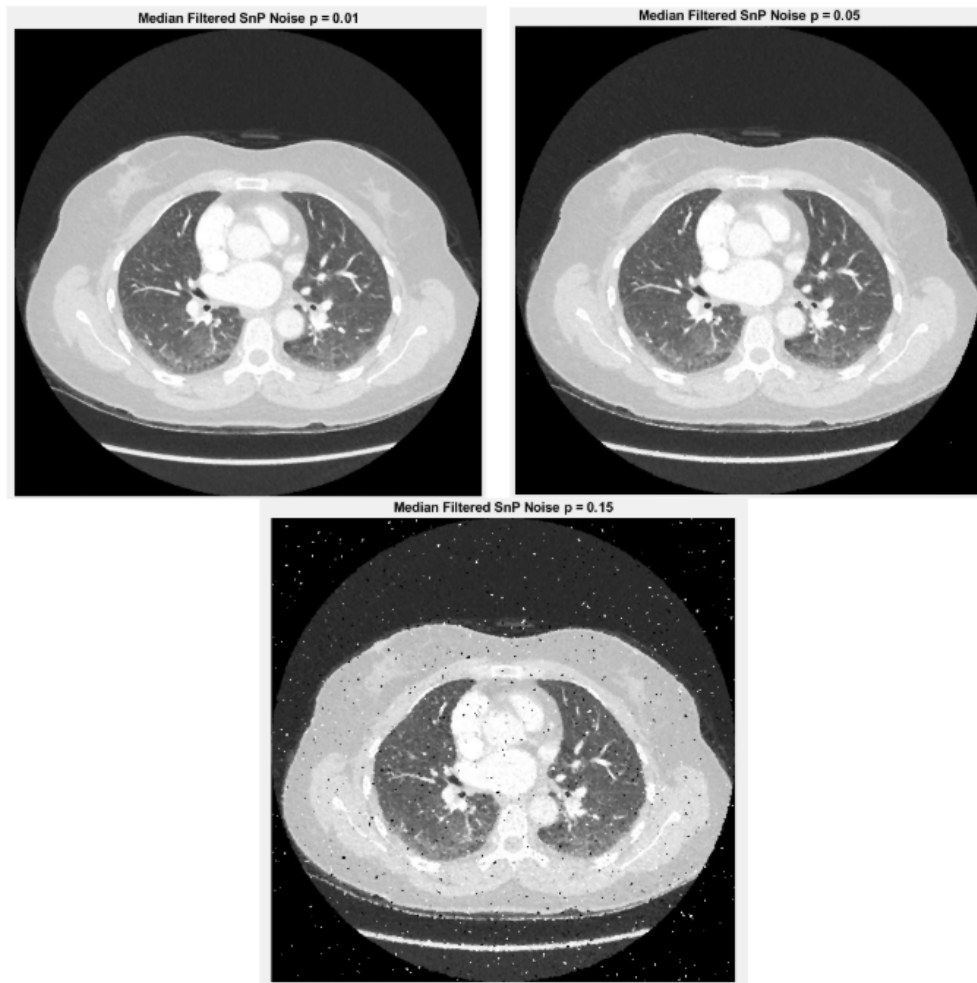


Figure 6: 3x3 Median Filtered Salt-N-Pepper Noise image slice $z = 130$.

Part III: Gauss' Wrath

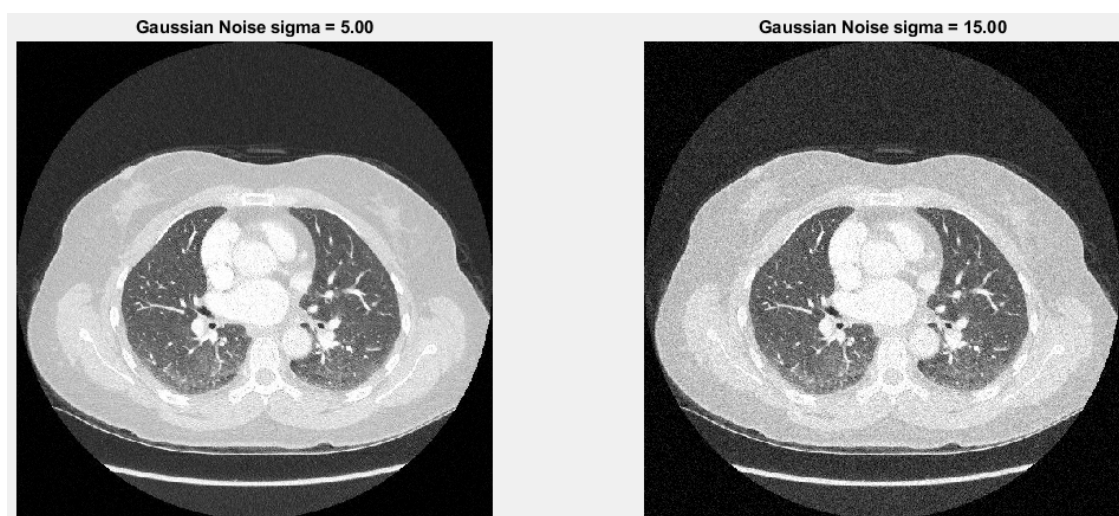


Figure 7: Gaussian Noise of $\sigma = 5$ and $\sigma = 15$ applied to slice $z = 130$.

Since Gaussian noise is present in most measurements due to hardware, Gaussian filters are typically employed to filter out this noise.

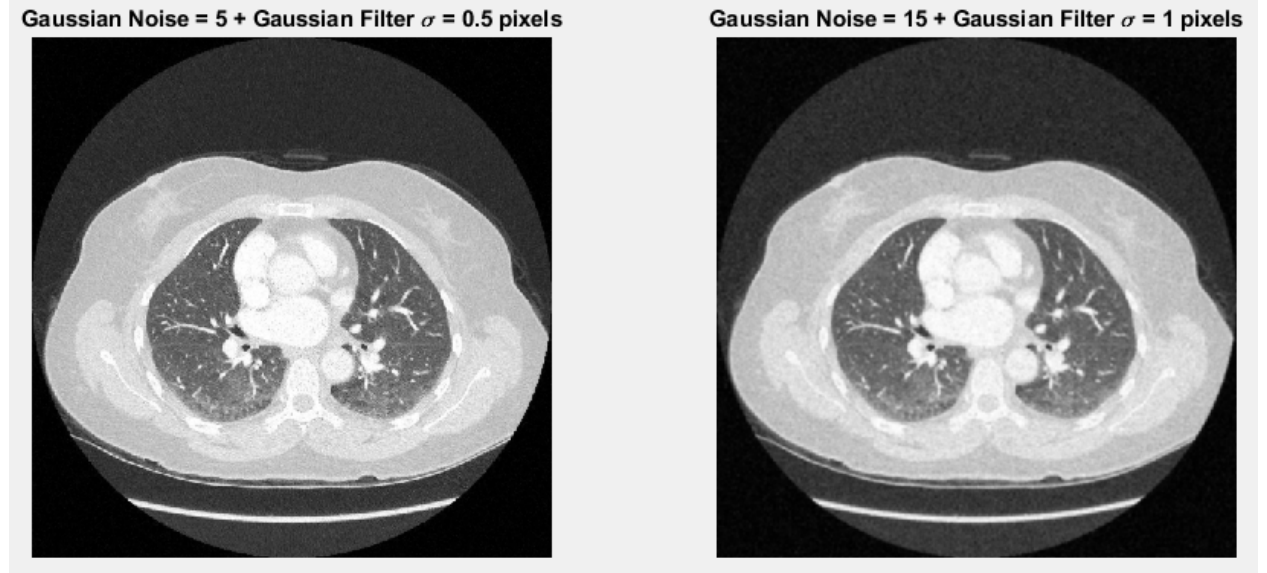


Figure 8: Gaussian Filter of $\sigma = 0.5$ pixels and $\sigma = 1$ pixel applied to noisy slice $z = 130$.

The additive Gaussian noise has been reduced, but the image is now noticeably blurrier than before. The larger the values of σ , the stronger smoothing but the higher degradation of edge sharpness. This effect is observed in the Gaussian filter with higher standard deviation $\sigma = 1$. Thus, there is an inherent tradeoff between noise reduction and image sharpness governed by the choice of σ .

Assuming that the CT scan is a perfect representation of the sample, this imaging system can be modeled using an anisotropic PSF. More specifically, this PSF can be modeled as a three-dimensional Gaussian pattern in space. The FWHM of a Gaussian PSF with variance σ is: $2\sqrt{2\ln(2)}\sigma \approx 2.35\sigma$. If the resolution quantified by the FWHM value of the PSF is δx in the lateral directions and δz in the axial, then the resulting Gaussian PSFs in each direction are:

$$\begin{aligned} x, y \text{ direction: PSF} &\sim \mathcal{N}\left(0, \left(\frac{\delta x}{2\sqrt{2\ln 2}}\right)^2\right) \\ z \text{ direction: PSF} &\sim \mathcal{N}\left(0, \left(\frac{\delta z}{2\sqrt{2\ln 2}}\right)^2\right) \end{aligned}$$

Since each of the x-y-z directions are independent, the 3-D Gaussian PSF is:

$$\text{PSF}(x, y, z) \sim \mathcal{MVN}(0, \Sigma), \quad \Sigma = \text{diag}\left(\frac{\delta x}{2\sqrt{2\ln 2}}, \frac{\delta x}{2\sqrt{2\ln 2}}, \frac{\delta z}{2\sqrt{2\ln 2}}\right)$$

Alternatively,

$$\begin{aligned}
\text{PSF}(x, y, z) &= \mathbf{C} \cdot \frac{1}{(\sqrt{2\pi})^3} \cdot \frac{1}{\sqrt{\det \Sigma}} \cdot \exp\left(-\frac{\mathbf{x}^\top \Sigma \mathbf{x}}{2}\right) \\
&= \mathbf{C} \cdot \frac{1}{(\sqrt{2\pi})^3} \cdot \frac{1}{\frac{\delta z}{2\sqrt{2\ln 2}} \left(\frac{\delta x}{2\sqrt{2\ln 2}}\right)^2} \cdot \exp\left(\frac{4\ln 2}{\delta x^2}x^2 + \frac{4\ln 2}{\delta x^2}y^2 + \frac{4\ln 2}{\delta z^2}z^2\right)
\end{aligned}$$

where $\mathbf{x} = [x \ y \ z]$ and $\mathbf{C}$ is a normalization constant which can be numerically computed.

***Note:** The general form is:

$$\text{PSF}(x, y, z) = \mathbf{C} \cdot \frac{1}{(2\pi)^{3/2} \sigma_x \sigma_y \sigma_z} \exp\left(-\left(\frac{x^2}{2\sigma_x^2} + \frac{y^2}{2\sigma_y^2} + \frac{z^2}{2\sigma_z^2}\right)\right)$$

Let $\delta x = 2\text{mm}$ and $\delta z = 1\text{mm}$. The resulting filtered images are:

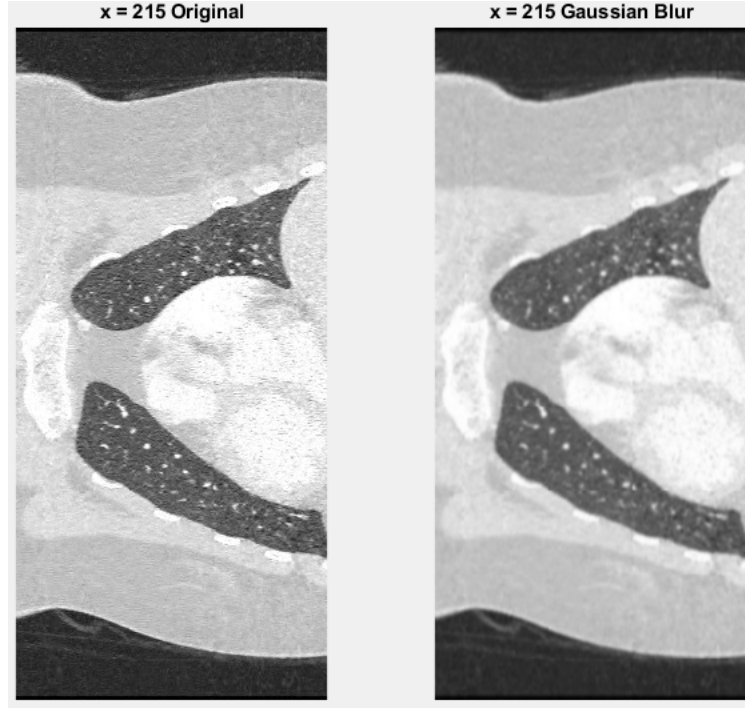


Figure 9: Gaussian Blurring applied to slice $x = 215$.

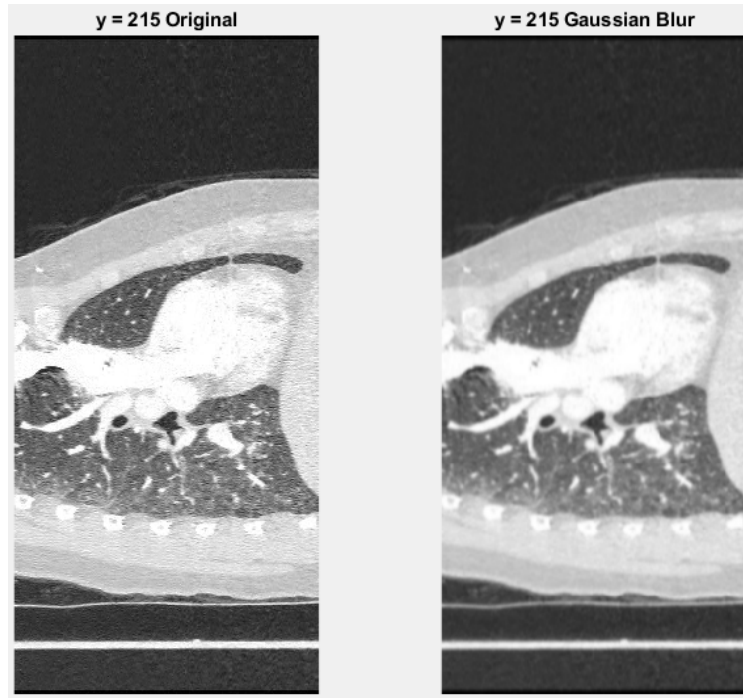


Figure 10: Gaussian Blurring applied to slice $y = 215$.

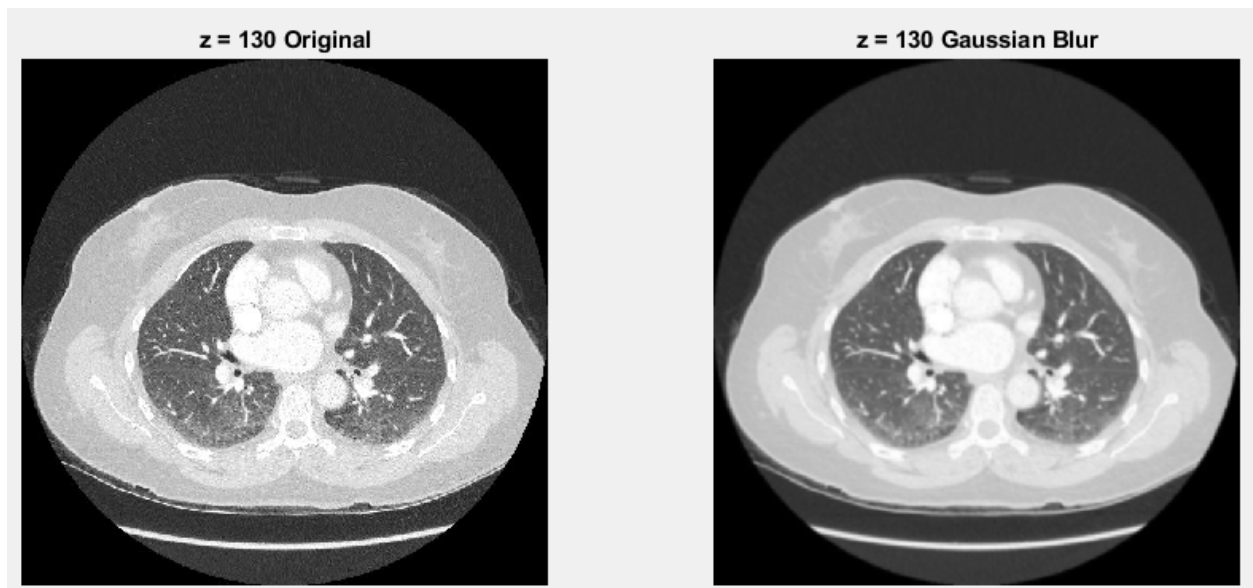


Figure 11: Gaussian Blurring applied to slice $z = 130$.

These blurred images look fairly impressive as the majority of the detail remains whilst the grainy textures are removed.

Part IV: Foramina

Looking at slice $y = 180$, the intervertebral foramina are visible (in between the white spots). When applying the Canny Edge Detection, the outline of the foramina becomes less visible. In fact, the foramina is only visible via the outline of the vertebrae, and even so it is not super clear since there are numerous spurious edges throughout the image.

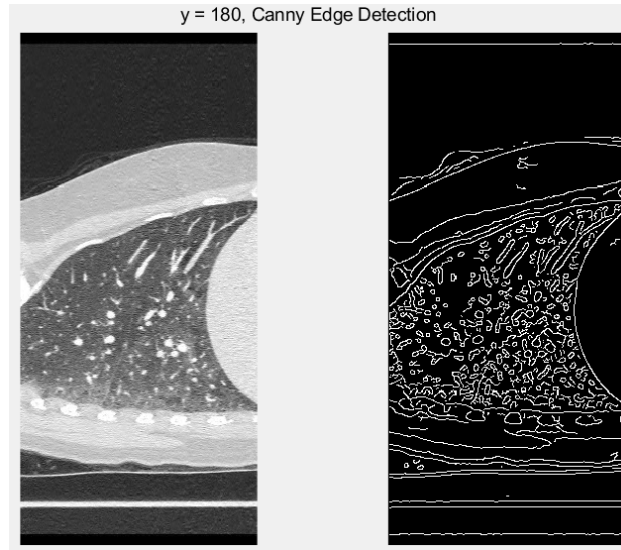


Figure 12: Canny Edge Detection applied to slice $y = 180$

Applying the Canny Edge Detection to 3x3 Median filtered image, there are some visible improvements. The filtered image removes some spurious edges in the raw Canny Edge image, and the vertebrae become more distinct and easier to see. However, there are still false edges throughout the center.

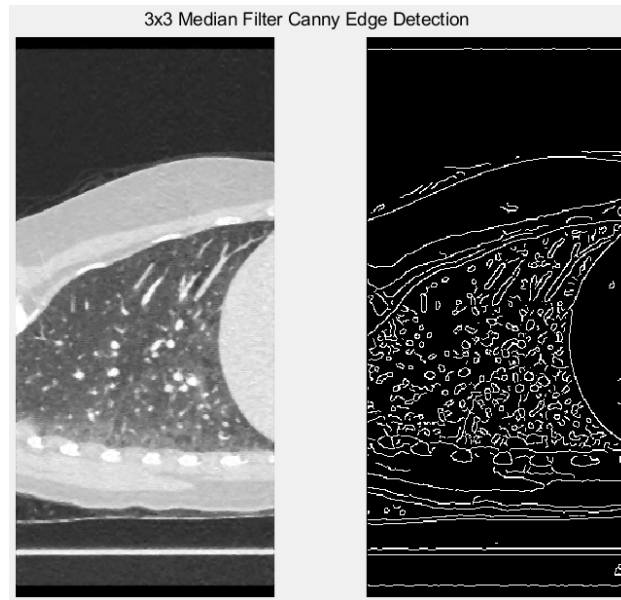


Figure 13: Canny Edge Detection applied to 3x3 Median filtered slice $y = 180$

Applying the Canny Edge Detection to the Gaussian filtered image described in section III, there again is a slight improvement compared to the 3x3 Median filtered image. In the top right corner, there are fewer false edges and the vertebrae disks (and the foramina) are more clear.

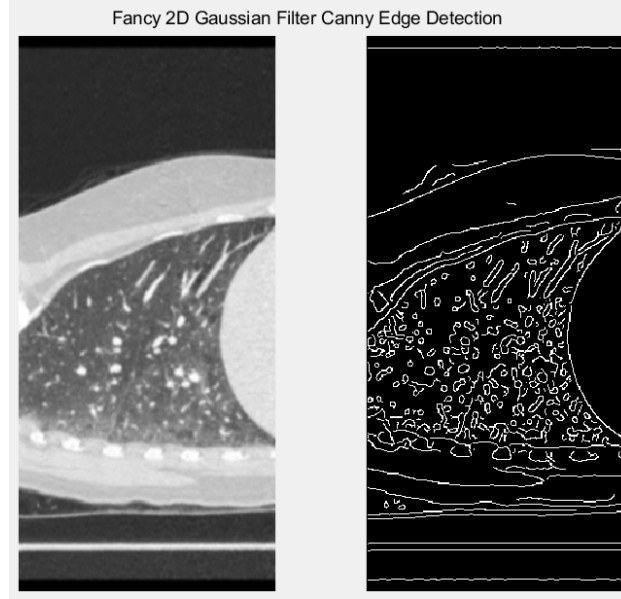


Figure 14: Canny Edge Detection applied to Fancy 2D Gaussian Filtered slice $y = 180$

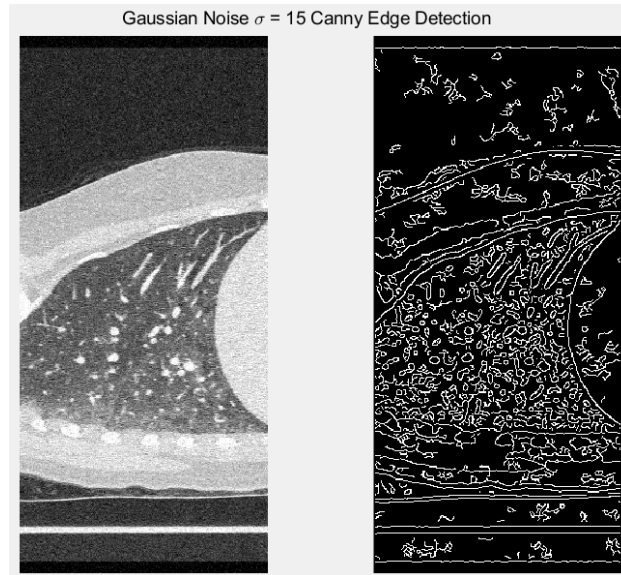


Figure 15: Canny Edge Detection applied to Gaussian noise $\sigma = 15$ slice $y = 180$

The Canny Edge Detection applied to the image added with Gaussian noise $\sigma = 15$ is the worst of the lot. Numerous spurious edges are outlined due to the addition of noise. The interior is severely degraded, making it difficult to differentiate true anatomical features from noise artifacts.

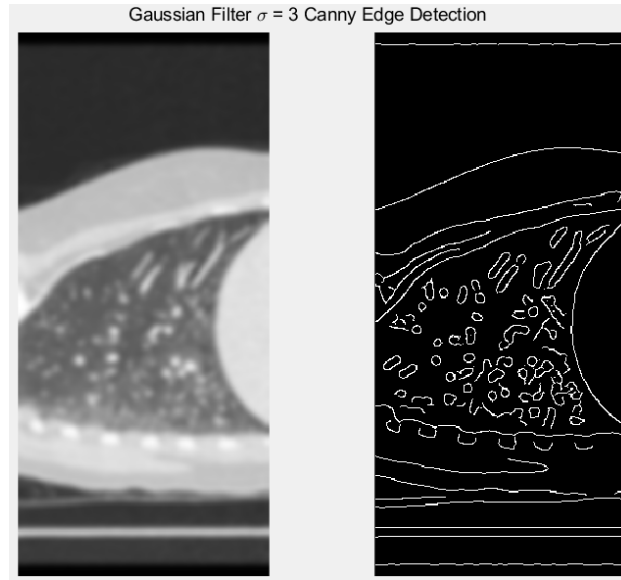


Figure 16: Canny Edge Detection applied to Gaussian filter $\sigma = 3$ pixels slice $y = 180$

Despite the initial image looking blurry, applying the Canny Edge Detection to the Gaussian filtered image produces an extremely clean result. There are almost no spurious edges outlined, and vertebrae, along with the foramina are clearly distinguishable.

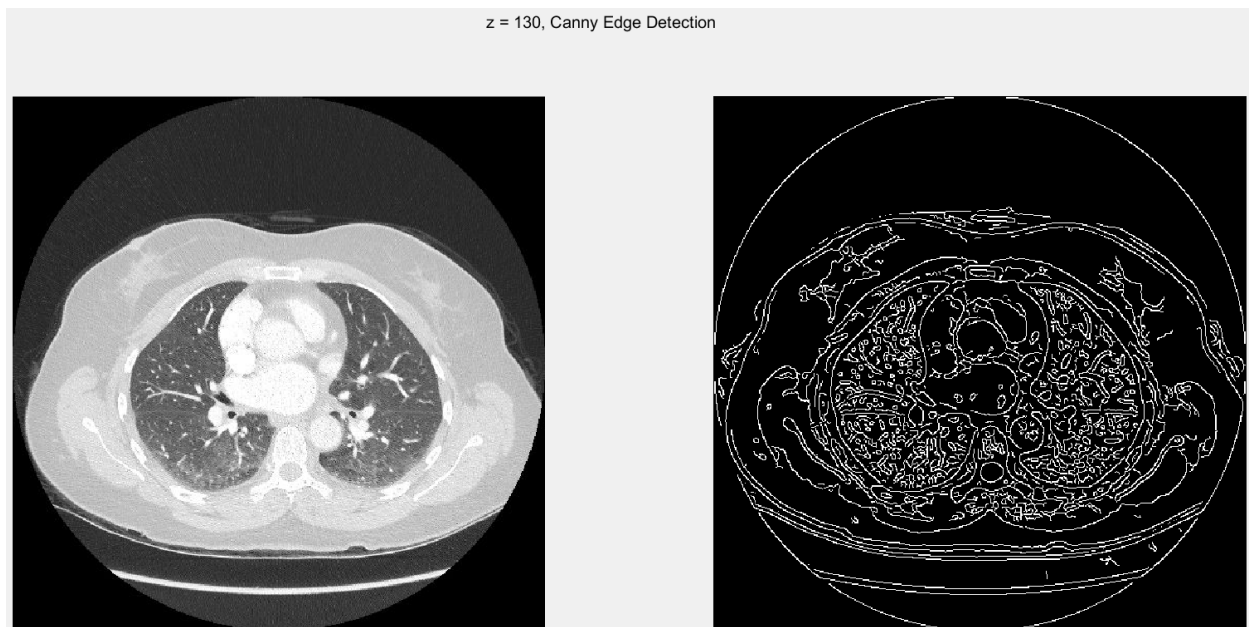


Figure 17: Canny Edge Detection applied to slice $z = 130$

Despite some spurious edges still present, the image is very nice and aesthetically pleasing!

Conclusion

This project demonstrates how noise characteristics and system resolution fundamentally influence medical image quality. Common noise problems such as Salt-n-pepper noise was shown to be effectively remediated through Median filtering, whilst Gaussian noise was shown to be remediated through Gaussian filtering. The effects of preprocessing images was especially highlighted when applying edge detection methods. Applying certain filters such as Gaussian blurring beforehand heavily influenced the quality of the edge detected images. This exploratory project highlights the critical role that image processing plays in field of of medical imaging.

References

- [1] J. Borgbjerg, “Mulrecon.” [Online]. Available:
<https://www.castlemountain.dk/atlas/index.php?page=mulrecon>